

7. [Maximum mark: 6]

Consider the following two equations.

$$2x + 6y - 8z = 13$$

$$3x - y + 3z = 12$$

- (a) Determine the general solution giving the answer in parametric form. [2]

Consider a third equation $ax + 12y - 16z = k$, where $a, k \in \mathbb{R}$.

- (b) In the case where $a = 3$ and $k = -3$, determine the unique solution to the system of three equations. [2]

- (c) (i) Write down the value of a for which there is no unique solution to the system of three equations.
- (ii) Hence, write down the corresponding value of k for which there is an infinite number of solutions to the system of three equations. [2]

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